

Protik Dey

Ph.D. Student, Department of Computer Science, University of Texas at San Antonio
San Antonio, Texas, USA

protik.dey@my.utsa.edu | [protikdey.github.io](https://github.com/protikdey) | [linkedin.com/in/protik-dey-59a881181](https://www.linkedin.com/in/protik-dey-59a881181)
scholar.google.com/citations?user=wj-omawAAAAJ | github.com/ProtikDey

SUMMARY

Human-Computer Interaction researcher studying how blind and low-vision (BLV) people use generative AI, and where these systems build trust or introduce new privacy and accessibility risks. Combines qualitative methods (interviews, user studies with BLV participants) with the design and evaluation of AI-based assistive prototypes. Previously a software engineer building financial-transaction systems with Spring Boot, Redis, and RabbitMQ.

RESEARCH AREA

Human-Computer Interaction, Trustworthy AI, Explainable AI, Usable Privacy, Accessibility

SKILLS

- **HCI & Research Methods:** User studies, thematic analysis, usability testing, interactive prototyping, accessible & inclusive design
- **Programming & Data:** Python, C, C++, Java, JavaScript, TypeScript, Bash, NumPy, Pandas, Scikit-learn, Matplotlib
- **Engineering Stack:** Spring Boot, Django, Angular, Redis, RabbitMQ, Kafka, Hibernate, Oracle, MySQL, PostgreSQL, MongoDB, Git

EDUCATION

Ph.D. in Computer Science

Aug 2024 – Present

University of Texas at San Antonio, Texas, USA

Advisor: [Dr. Taslima Akter](#) | [APEX Lab](#)

B.Sc. (Engg.) in Computer Science and Engineering

Feb 2016 – Mar 2021

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Thesis: *Malicious Code Detection and Analysis*, advised by Dr. Muhammad Masroor Ali

PUBLICATIONS

Peer-Reviewed Papers

1. Syed Zami-Ul-Haque Navid, **Protik Dey**, Shamiul Hasan, and Muhammad Masroor Ali, “Static Detection of Malicious Code in Programs Using Semantic Techniques”, In the Proceedings of the 11th International Conference on Electrical and Computer Engineering (**ICECE 2020**).

Workshop Papers

1. **Protik Dey**, Mohd. Saifuzzaman, and Taslima Akter, “More Than Just Access: Generative AI as Communication Intermediary for Blind and Low-Vision Users”, **ACM CSCW 2026 Workshop on Broader Impacts of GenAI in Communication: Building Agendas for Research, Design, and Policymaking**.
2. Abu Noman Md Sakib, **Protik Dey**, Zijie Zhang, and Taslima Akter, “Explainable AI for Blind and Low-Vision Users: Navigating Trust, Modality, and Interpretability in the Agentic Era”, ACM CHI Workshop on Human-Centered Explainable AI (**HCTXAI 2026**).

TEACHING EXPERIENCE

Graduate Teaching Assistant

Aug 2024 – Dec 2025

University of Texas at San Antonio, Texas, USA

- Teaching assistant for undergraduate Operating Systems and Systems Programming.
- Graded assignments, projects, and exams; supported students through office hours on OS concepts and low-level C programming.

INDUSTRY EXPERIENCE

Software Engineer II, Kona Software Lab Limited

Jan 2023 – Jul 2024

Integrated Platform Team, Dhaka, Bangladesh

- Worked in the Integrated Platform team building global payment solutions.
- Designed and implemented validation and authorization for financial transactions, including recharge and payment flows.
- *Tech Stack: Spring Boot, Redis, RabbitMQ, MariaDB*

Graduate Engineer, Kona Software Lab Limited

Aug 2021 – Dec 2022

Digital Financial System Team, Dhaka, Bangladesh

- Full-stack developer on the Digital Financial System team, delivering financial transaction solutions for Nagad, Bangladesh's fastest-growing mobile financial system, including tokenized payments, KYC, and bank/card transfers.
- Helped stand up the back-office portal for the KONAPLATE project using Angular.
- *Tech Stack: Spring Boot, Angular, Redis, RabbitMQ, Strapi, Oracle*

CURRENT ROLE

Graduate Research Assistant II

Jan 2026 – Present

University of Texas at San Antonio, Texas, USA

- Lead research on accessible, trustworthy, and privacy-aware generative AI for blind and low-vision (BLV) users.
- Design and conduct user studies with BLV participants; build research prototypes to evaluate accessible AI interactions, from study design and interviews through qualitative analysis and publication.
- Mentor undergraduate students on research methods and academic writing.

SELECTED RESEARCH PROJECTS

1. **Trustworthy Generative AI for Blind and Low-Vision Users:** Study how BLV people use generative AI as a communication intermediary and where trust, modality, interpretability, and privacy break down, using interviews and user studies alongside AI-based assistive prototypes.
2. **Fake Bengali Facebook Post Detection on COVID-19:** Built a benchmark dataset of Bengali-language Facebook posts on COVID-19 collected via CrowdTangle, trained transformer-based models on it, and reported on the correlation between misinformation and audience engagement, one of the first empirical studies of Bengali-language misinformation during the pandemic.
3. **Malicious Code Detection and Analysis (B.Sc. Thesis):** Extended CodeOntology to express the semantic structure of Java source code as an ontology encoding malware signatures (e.g., repeated connection calls behind a DDoS attack). Identified a known malware from source code alone with human-interpretable explanations, robust to obfuscation that scatters malicious calls across intermediate methods.
4. **Highly Dense Terahertz Communication Networks:** Simulated 5G and beyond-5G communication in NS-3 and analyzed throughput and scaling behavior of highly dense terahertz networks.

Contributed as External Reviewer (CHI 2026, ASSETS 2025, CSCW 2025 and 2026); volunteered at a BLV community event, Vibrant Works, San Antonio (April 2026).